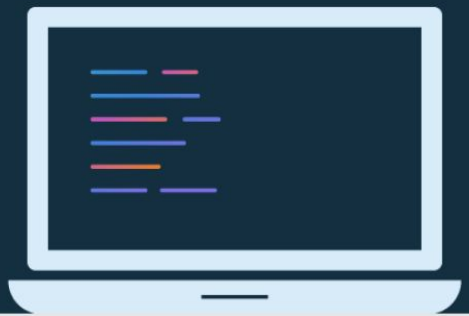


Lesson 4: Build your first Android app



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About this lesson

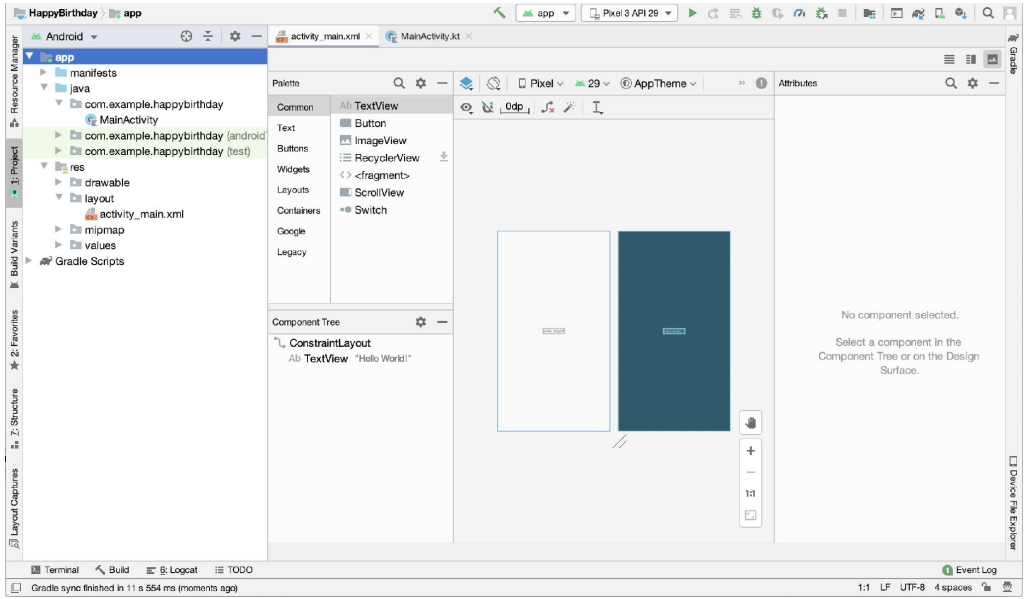
Lesson 4: Build your first Android app

- Your first app
- Anatomy of an Android app
- Layouts and resources in Android
- Activities
- Make an app interactive
- Gradle: Building an Android app
- Accessibility
- Summary

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Android Studio

Official IDE for building Android apps

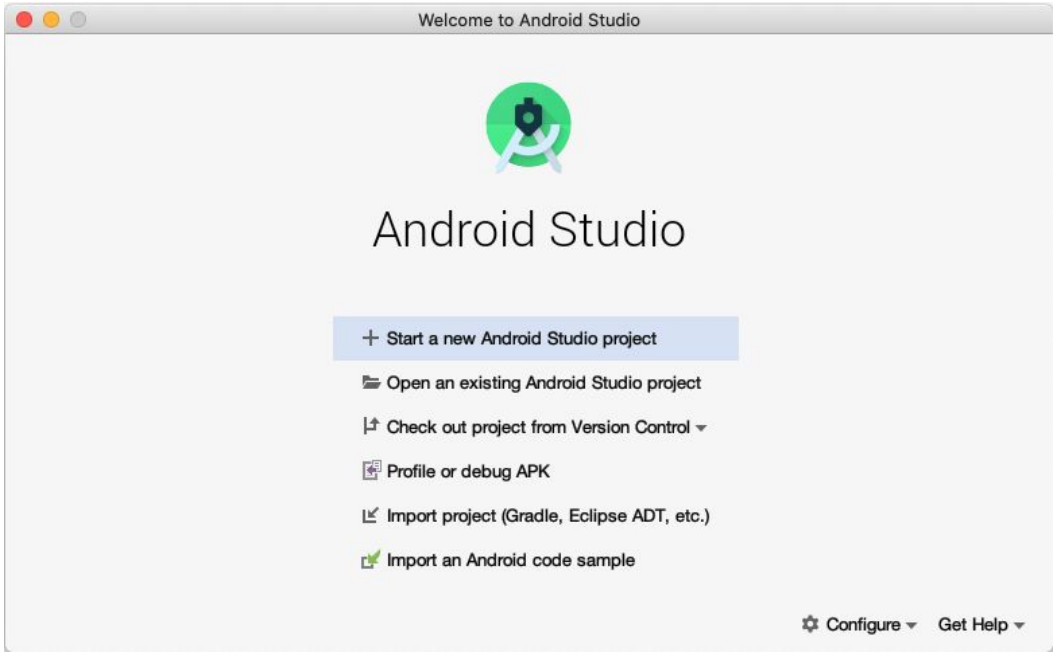


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Your first app

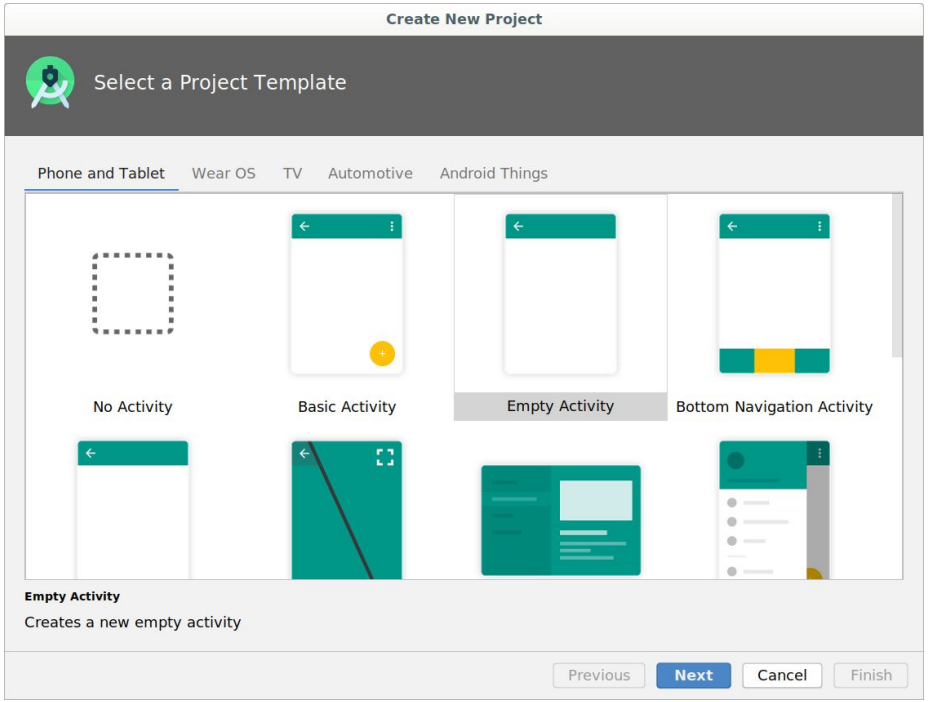
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Open Android Studio



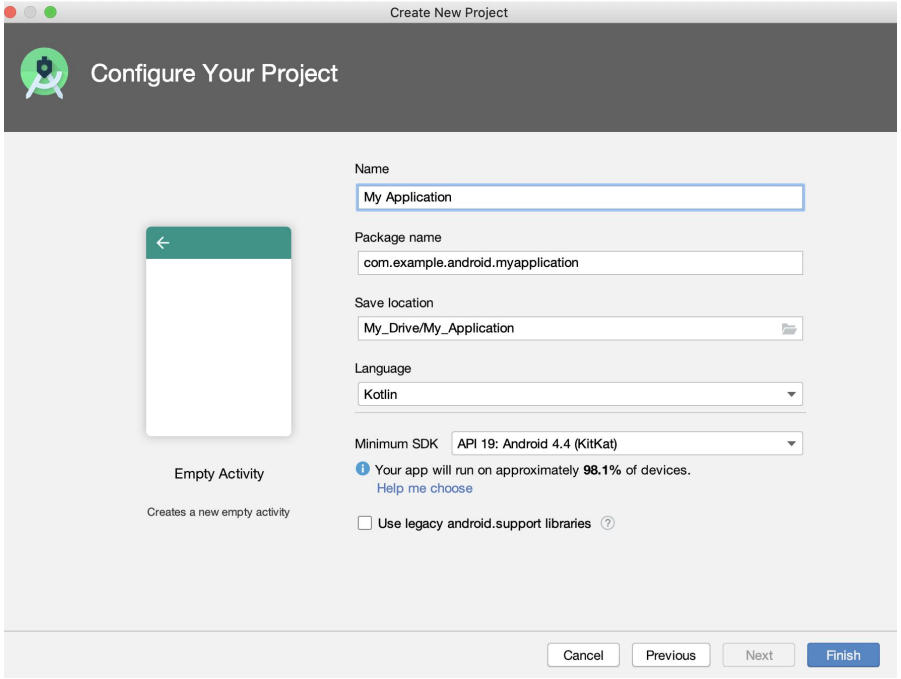
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Create new project



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Enter your project details



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Android releases and API levels

Platform Version	API Level	VERSION_CODE
Android 10.0	29	Q
Android 9	28	P
Android 8.1	27	O_MR1
Android 8.0	26	O
Android 7.1.1 Android 7.1	25	N_MR1
Android 7.0	24	N
Android 6.0	23	M
Android 5.1	22	LOLLIPOP_MR1
Android 5.0	21	LOLLIPOP

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Choose API levels for your app

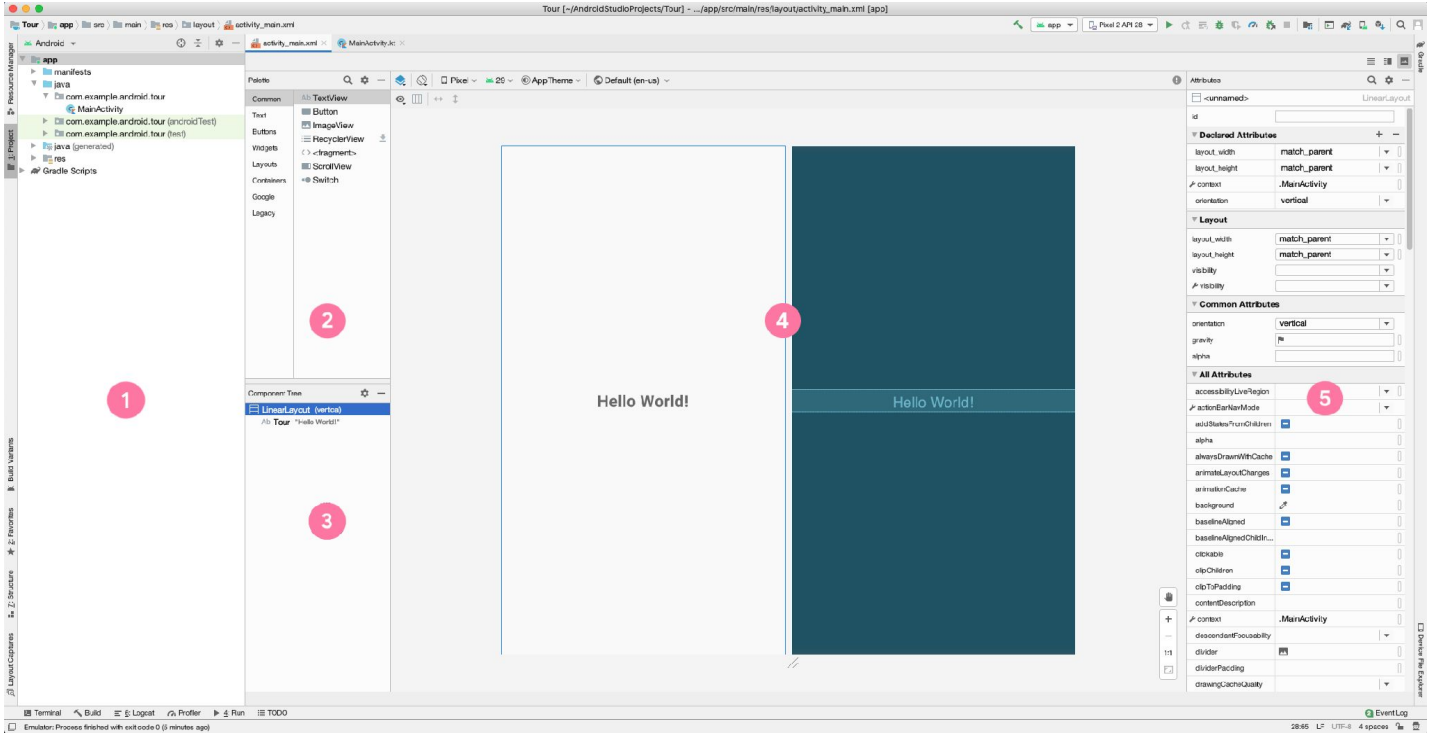
- Minimum SDK: Device needs at least this API level to install
- Target SDK: API version and highest Android version tested
- Compile SDK: Android OS library version compiled with

```
minSdkVersion <= targetSdkVersion <= compileSdkVersion
```

The API level identifies the framework API version of the Android SDK.

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Tour of Android Studio



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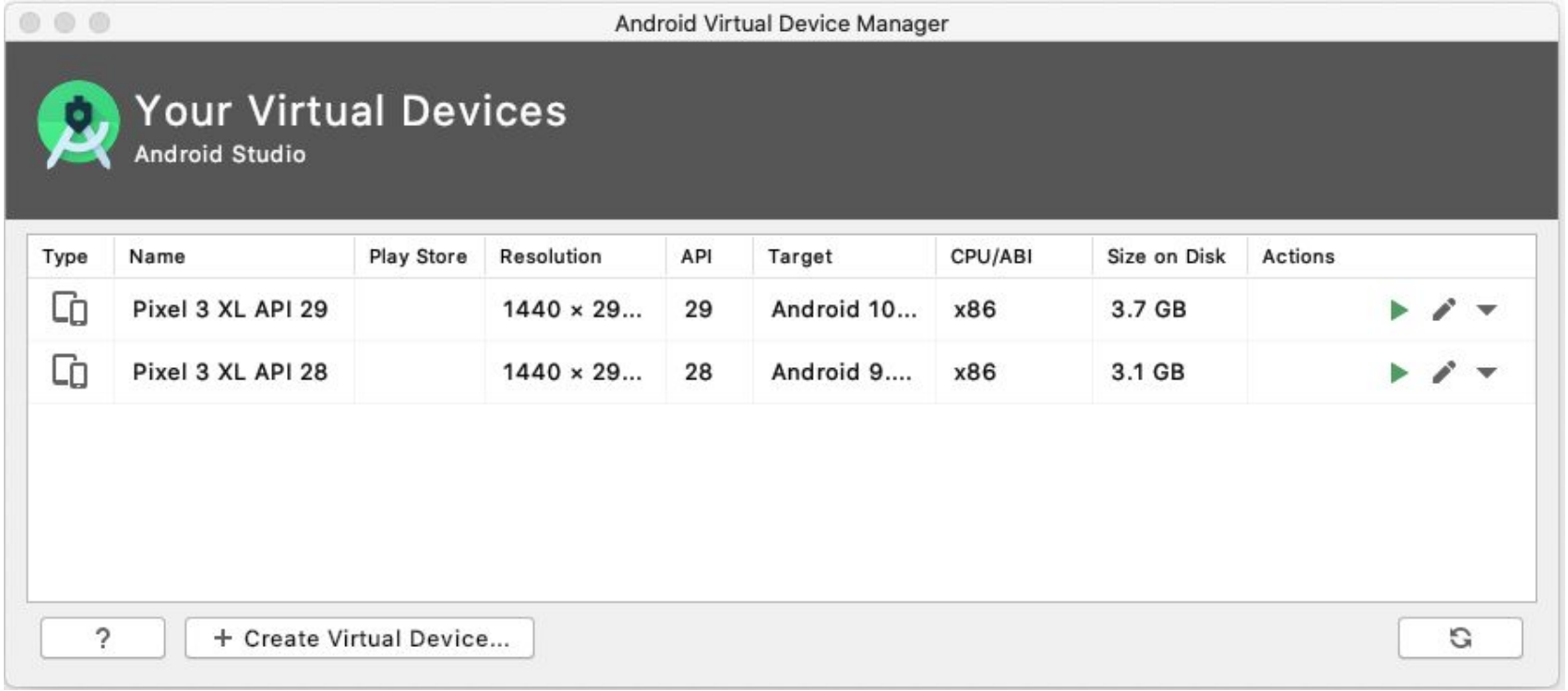
Run your app



- Android device (phone, tablet)
- Emulator on your computer

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Android Virtual Device (AVD) Manager



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Anatomy of an Android App project

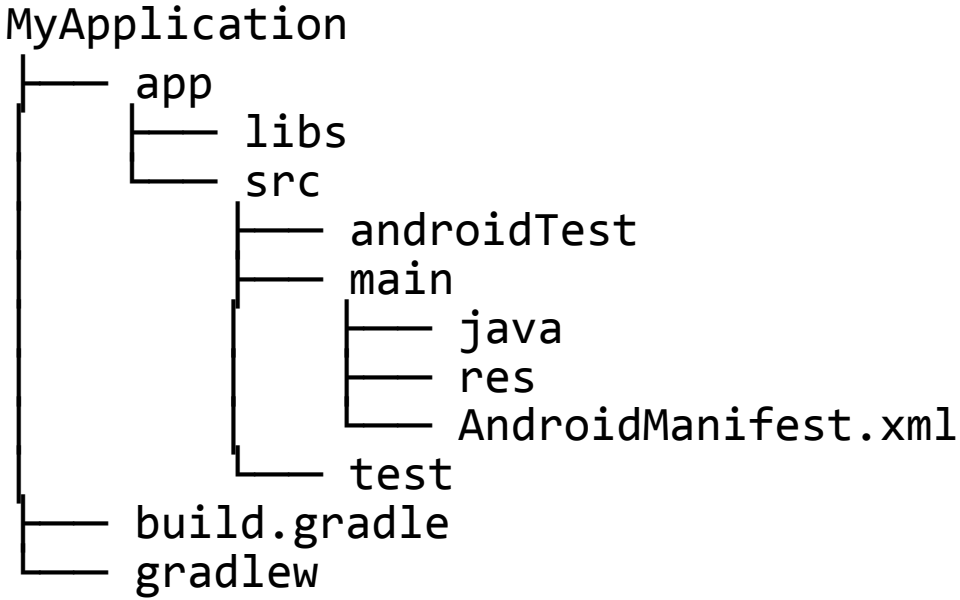
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Anatomy of a basic app project

- Activity
- Resources (layout files, images, audio files, themes, and colors)
- Gradle files

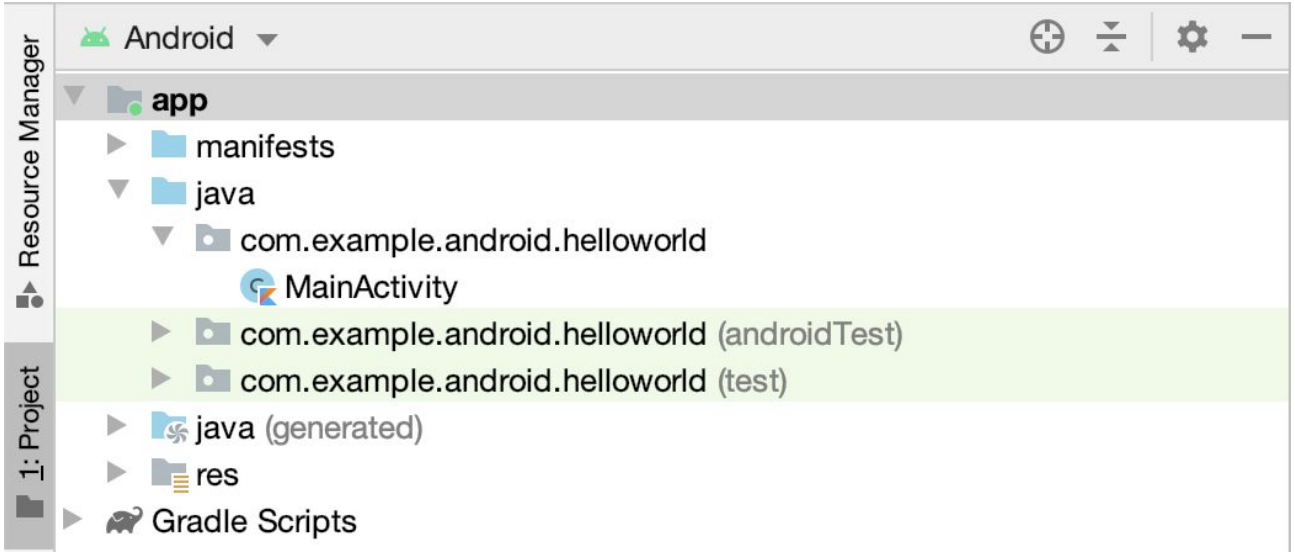
This work is licensed under the Apache 2 license.

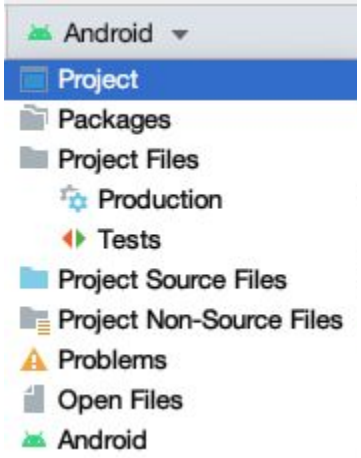
Android app project structure



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Browse files in Android Studio





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Layouts and resources in Android

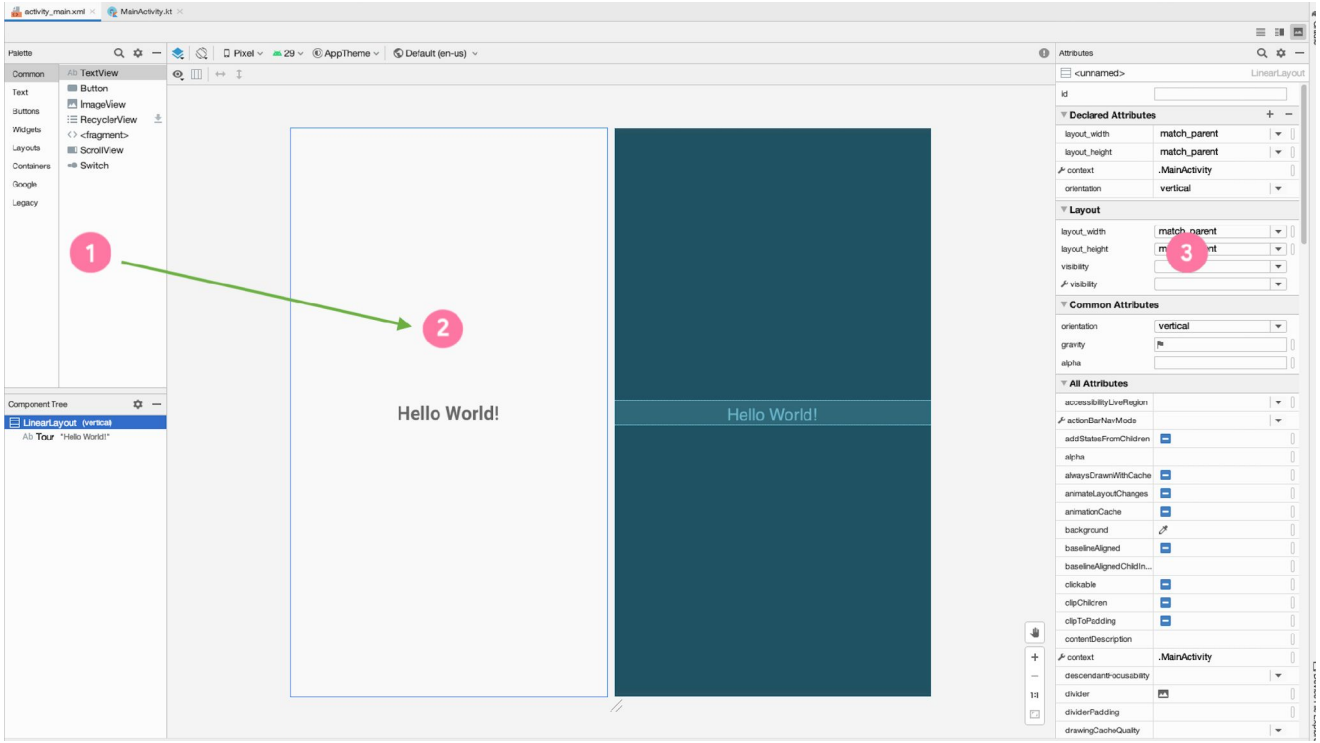
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Views

- Views are the user interface building blocks in Android
  - Bounded by a rectangular area on the screen
  - Responsible for drawing and event handling
  - Examples: TextView, ImageView, Button
- Can be grouped to form more complex user interfaces

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Layout Editor



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XML Layouts

You can also edit your layout in XML.

- Android uses XML to specify the layout of user interfaces (including View attributes)

- Each View in XML corresponds to a class in Kotlin that controls how that View functions

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XML for a TextView

```
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Hello World!"/>
```



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Size of a View

- wrap_content

```
android:layout_width="wrap_content"
```

- match_parent

```
android:layout_width="match_parent"
```

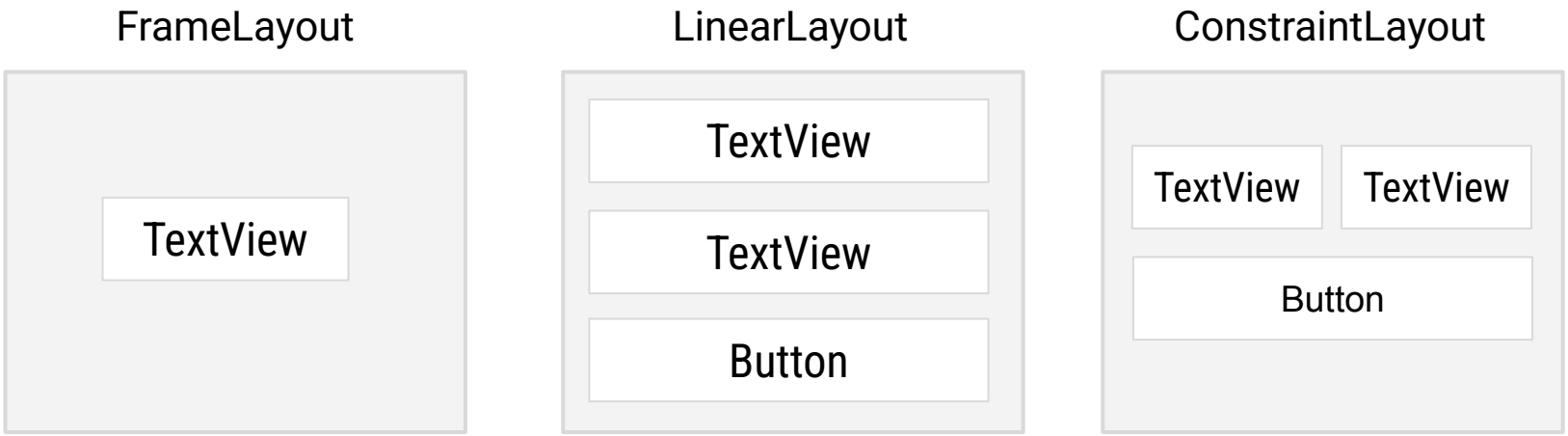
- Fixed value (use dp units)

```
android:layout_width="48dp"
```


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ViewGroups

A `ViewGroup` is a container that determines how views are displayed.



The ViewGroup is the parent and the views inside it are its children.

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FrameLayout example

A `FrameLayout` generally holds a single child `View`.

```
<FrameLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:text="Hello World!"/>
</FrameLayout>
```

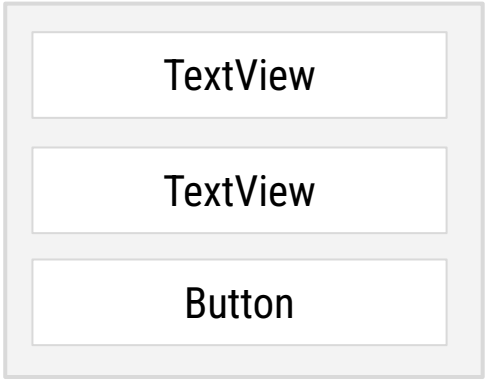


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LinearLayout example

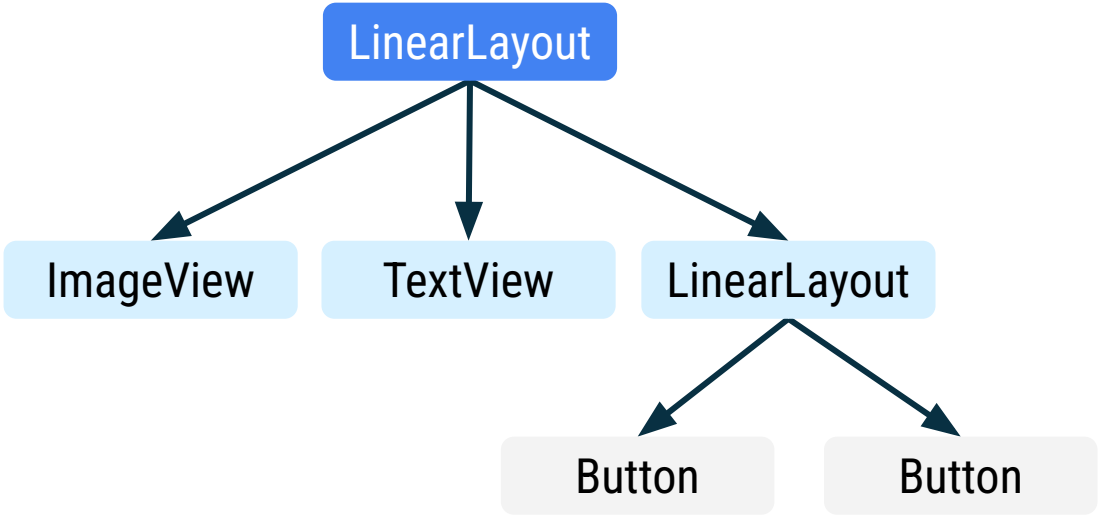
- Aligns child views in a row or column
- Set `android:orientation` to `horizontal` or `vertical`

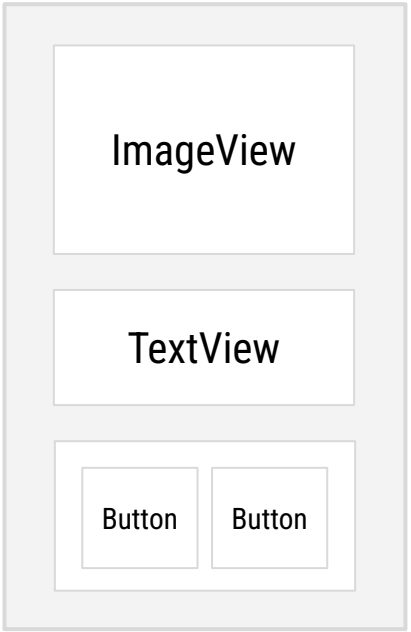
```
<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">
    <TextView ... />
    <TextView ... />
    <Button ... />
</LinearLayout>
```



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View hierarchy





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App resources

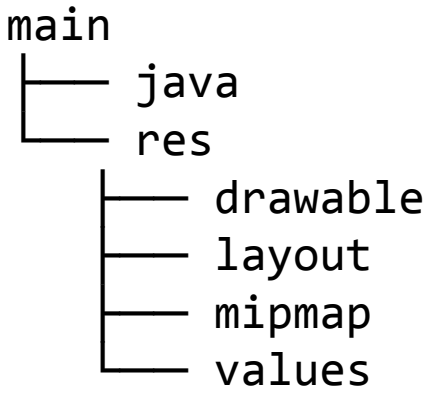
Static content or additional files that your code uses

- Layout files
- Images
- Audio files
- User interface strings
- App icon

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Common resource directories

Add resources to your app by including them in the appropriate resource directory under the parent `res` folder.



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Resource IDs

- Each resource has a resource ID to access it.
- When naming resources, the convention is to use all lowercase with underscores (for example, `activity_main.xml`).
- Android autogenerates a class file named `R.java` with references to all resources in the app.
- Individual items are referenced with:
  `R.<resource_type>.<resource_name>`

Examples:
```
R.drawable.ic_launcher (res/drawable/ic_launcher.xml)
R.layout.activity_main (res/layout/activity_main.xml)
```


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Resource IDs for views

Individual views can also have resource IDs.

Add the `android:id` attribute to the View in XML. Use `@+id/name` syntax.

```
<TextView
    android:id="@+id/helloTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Hello World!"/>
```

Within your app, you can now refer to this specific TextView using:

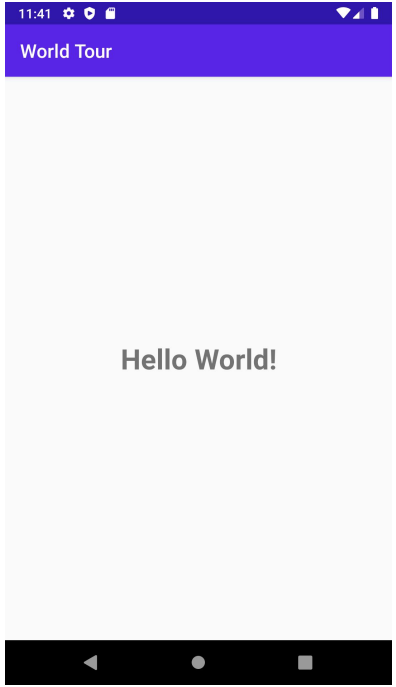
`R.id.helloTextView`

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Activities

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What's an Activity?



- An Activity is a means for the user to accomplish one main goal.
- An Android app is composed of one or more activities.

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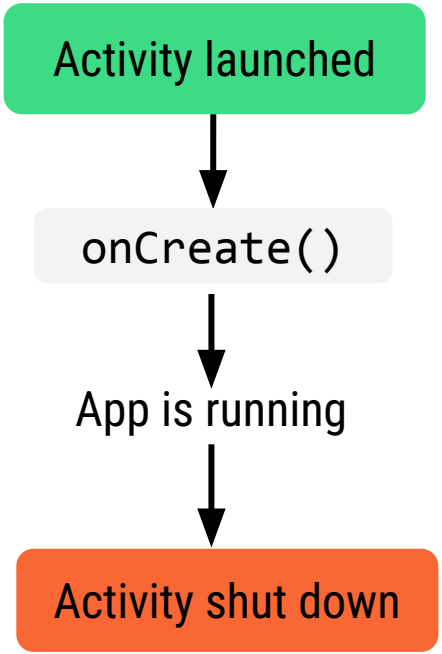
MainActivity.kt

```
class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)
    }
}
```


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How an Activity runs



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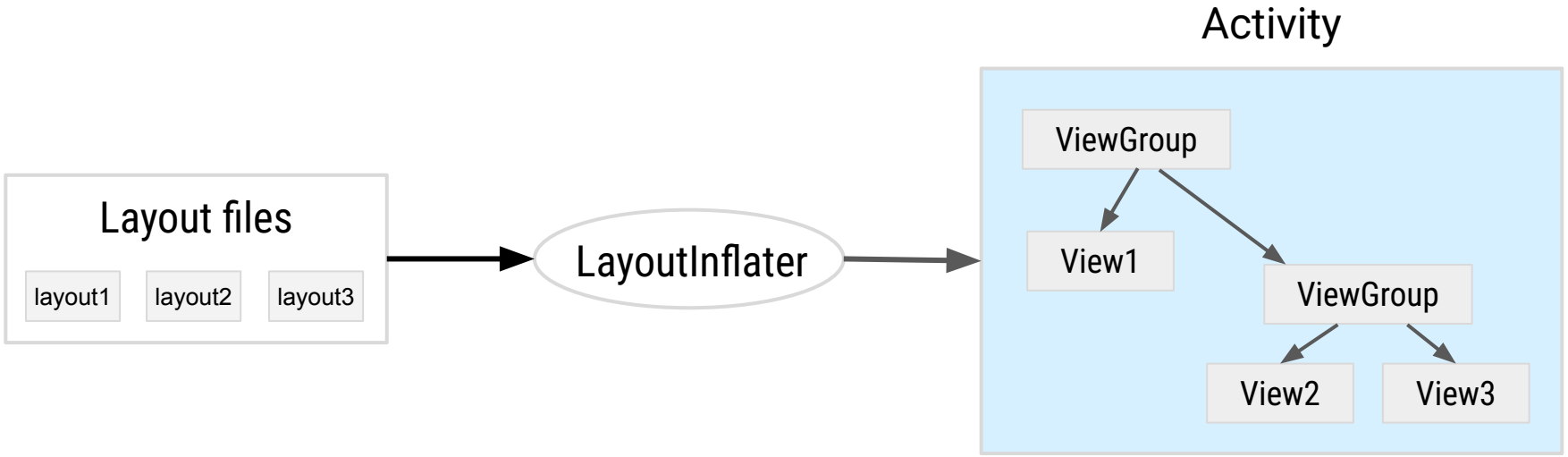
Implement the onCreate() callback

Called when the system creates your Activity

```
override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)
}
```


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Layout inflation



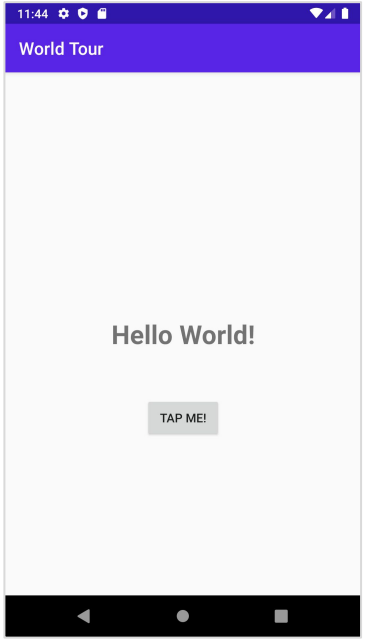
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Make an app interactive

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Define app behavior in Activity

Modify the Activity so the app responds to user input, such as a button tap.



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Modify a View dynamically

Within `MainActivity.kt`:

Get a reference to the View in the view hierarchy:

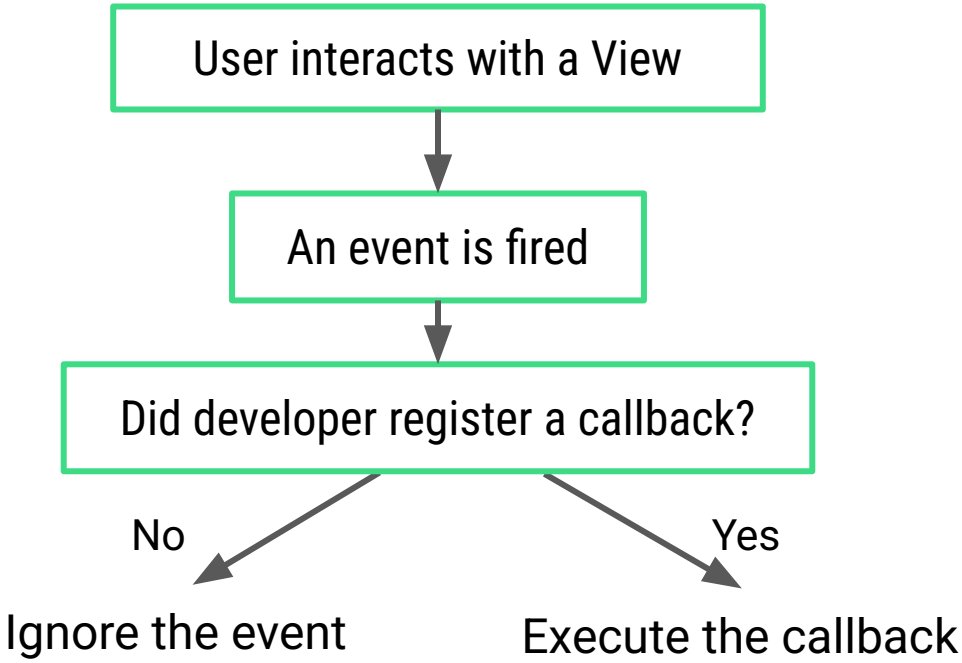
```
val resultTextView: TextView = findViewById(R.id.textView)
```

Change properties or call methods on the View instance:

```
resultTextView.text = "Goodbye!"
```


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Set up listeners for specific events



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View.OnClickListener

```
class MainActivity : AppCompatActivity(), View.OnClickListener {

    override fun onCreate(savedInstanceState: Bundle?) {
        ...
        val button: Button = findViewById(R.id.button)
        button.setOnClickListener(this)
    }

    override fun onClick(v: View?) {
        TODO("not implemented")
    }
}
```


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SAM (single abstract method)

Converts a function into an implementation of an interface

Format: `InterfaceName { lambda body }`

```
val runnable = Runnable { println("Hi there") }
```

is equivalent to

```
val runnable = (object: Runnable {
    override fun run() {
        println("Hi there")
    }
})
```


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View.OnClickListener as a SAM

A more concise way to declare a click listener

```
class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        ...

        val button: Button = findViewById(R.id.button)
        button.setOnClickListener({ view -> /* do something*/ })
    }
}
```


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Late initialization

```
class Student(val id: String) {

    lateinit var records: HashSet<Any>

    init {
        // retrieve records given an id
    }
}
```


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Lateinit example in Activity

```
class MainActivity : AppCompatActivity() {

    lateinit var result: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        ...
        result = findViewById(R.id.result_text_view)
    }
}
```


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Gradle: Building an Android app

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What is Gradle?

- Builds automation system
- Manages the build cycle via a series of tasks (for example, compiles Kotlin sources, runs tests, installs app to device)
- Determines the proper order of tasks to run
- Manages dependencies between projects and third-party libraries

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Gradle build file

- Declare plugins
- Define Android properties
- Handle dependencies
- Connect to repositories

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Plugins

Provide libraries and infrastructure needed by your app

```
apply plugin: 'com.android.application'

apply plugin: 'kotlin-android'

apply plugin: 'kotlin-android-extensions'
```


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Android configuration

```
android {
    compileSdkVersion 30
    buildToolsVersion "30.0.2"

    defaultConfig {
        applicationId "com.example.sample"
        minSdkVersion 19
        targetSdkVersion 30
    }
}
```


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Dependencies

```
dependencies {
    implementation "org.jetbrains.kotlin:kotlin-stdlib-jdk7:$kotlin_version"
    implementation 'androidx.core:core-ktx:1.3.2'
    implementation 'androidx.appcompat:appcompat:1.2.0'
    implementation 'com.google.android.material:material:1.2.1'
    ...
}
```


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Repositories

```
repositories {
    google()
    jcenter()
    maven {
        url "https://maven.example.com"
    }
}
```


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Common Gradle tasks

- Clean
- Tasks
- InstallDebug

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Accessibility

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Accessibility

- Refers to improving the design and functionality of your app to make it easier for more people, including those with disabilities, to use
- Making your app more accessible leads to an overall better user experience and benefits all your users

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Make apps more accessible

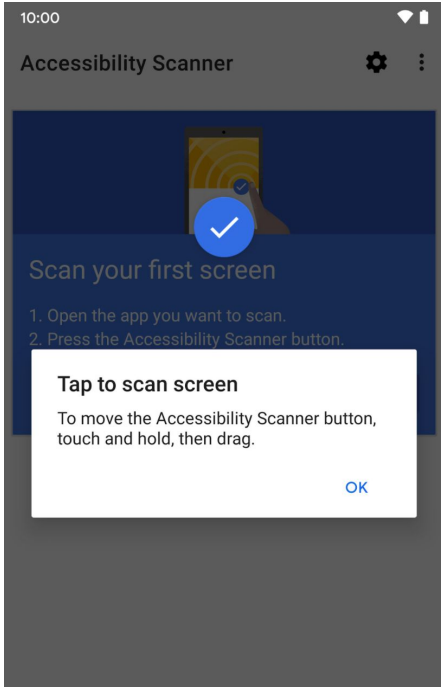
- Increase text visibility with foreground and background color contrast ratio:
  - At least 4.5:1 for small text against the background
  - At least 3.0:1 for large text against the background
- Use large, simple controls
  - Touch target size should be at least 48dp x 48dp
- Describe each UI element
  - Set content description on images and controls

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Accessibility Scanner

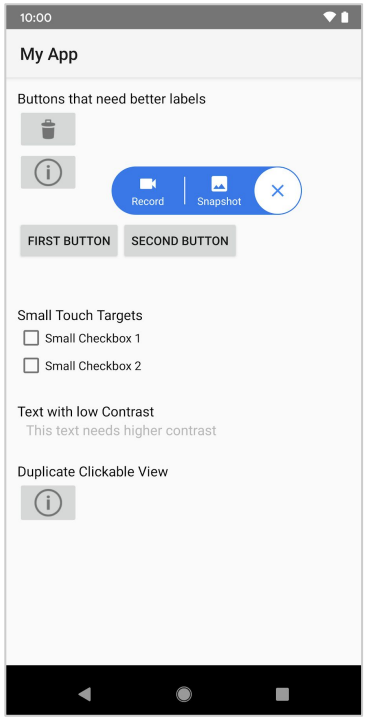
Tool that scans your screen and suggests improvements to make your app more accessible, based on:

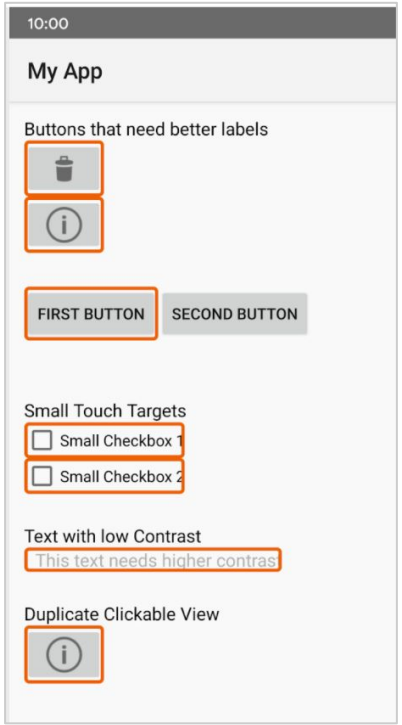
- Content labels
- Touch target sizes
- Clickable views
- Text and image contrast

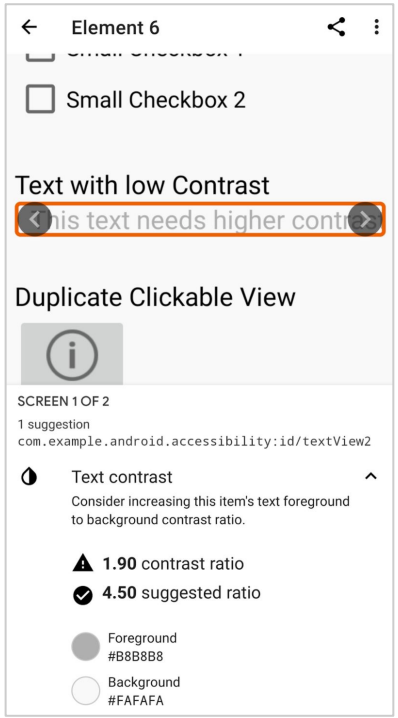


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Accessibility Scanner example







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Add content labels

- Set `contentDescription` attribute → read aloud by screen reader

```
<ImageView
    ...
    android:contentDescription="@string/stop_sign" />
```

- Text in TextView already provided to accessibility services, no additional label needed

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No content label needed

- For graphical elements that are purely for decorative purposes, you can set

```
android:importantForAccessibility="no"
```

- Removing unnecessary announcements is better for the user

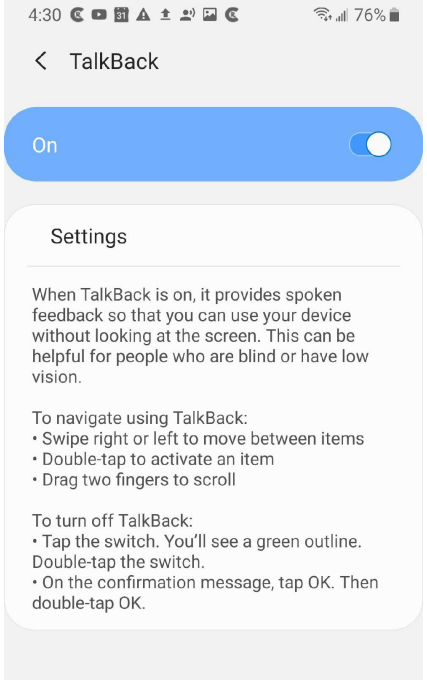
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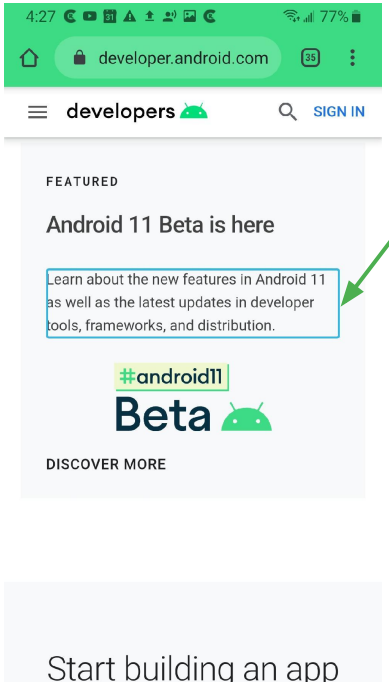
TalkBack

- Google screen reader included on Android devices
- Provides spoken feedback so you don’t have to look at the screen to use your device
- Lets you navigate the device using gestures
- Includes braille keyboard for Unified English Braille

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TalkBack example





Reads text aloud as user navigates the screen

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Switch access

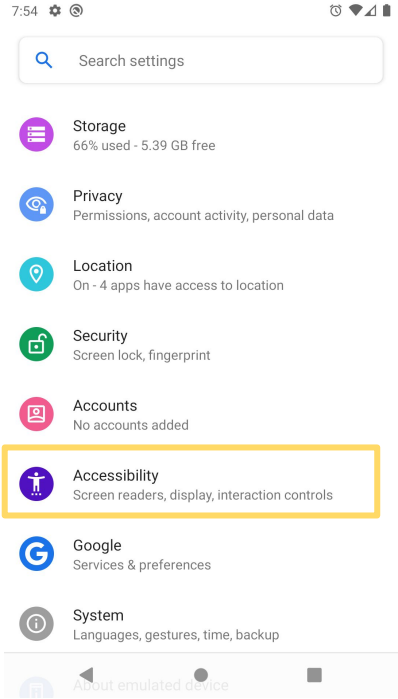
- Allows for controlling the device using one or more switches instead of the touchscreen
- Scans your app UI and highlights each item until you make a selection
- Use with external switch, external keyboard, or buttons on the Android device (e.g., volume buttons)

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Android Accessibility Suite

Collection of accessibility apps that help you use your Android device eyes-free, or with a switch device. It includes:

- Talkback screen reader
- Switch Access
- Accessibility Menu
- Select to Speak



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Accessibility Resources

- Build more accessible apps
- Principles for improving app accessibility
- Basic Android Accessibility codelab
- Material Design best practices on accessibility

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Summary

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Summary

In Lesson 4, you learned how to:

- Use Views and `ViewGroups` to build the user interface of your app
- Access resources in your app from `R.<resource_type>.<resource_name>`
- Define app behavior in the Activity (for example, register `OnClickListener`)
- Use Gradle as the build system to build your app
- Follow best practices to make your apps more accessible

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Learn more

- Layouts
- LinearLayout
- Input events overview
- View
- ViewGroup

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Pathway

Practice what you've learned by completing the pathway:

Lesson 4: Build your first Android app



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